Supplementary Information

A Synthetic Chemoinformatics Testbed

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S1. Substrate Library Mapping

Exact monoisotopic masses (`ExactMolWt`, in Da) were computed using RDKit (`Descriptors.ExactMolWt`) directly from the 22 substrate SMILES strings and rounded to two decimal places.

Substrate_Idx	SMILES	ExactMolWt (Da)
0	CC(C)O	60.06
1	CCC(C)O	74.07
2	CCCC(C)O	88.09
3	CC(C)C(C)O	88.09
4	C1CCCCC1O	100.09
5	CC1CCCCC1O	114.10
6	CC1CCC(C(C)C)CC1O	156.15
7	c1ccccc1C(C)O	122.07
8	c1ccccc1CC(C)O	136.09
9	COc1ccccc1C(C)O	152.08
10	Clc1ccccc1C(C)O	156.03
11	CC(=O)c1ccc(C(C)O)cc1	164.08
12	C1=CC=C2C(=C1)C=CC(=C2)C(C)O	172.09
13	CC1=CC=C(C=C1)C(C)O	136.09
14	c1ccc2c(c1)Cc3ccccc3C2O	196.09
15	c1cc2cc3c(cc2cc1)CC(O)C3	184.09
16	CC(C)(C)c1ccc(C(C)O)cc1	178.14
17	C1CCC(CC1)C(C)O	128.12
18	C1=CC=C(C=C1)C(O)CC	136.09
19	c1ccc(cc1)C(c2ccccc2)O	184.09
20	C1CCC(CC1)C(c2ccccc2)O	190.14
21	c1ccc2c(c1)CCc3ccccc3C2O	210.10

S2. Categorical Parameter Indices and Sampling Probabilities

Substrate_Idx: Sampled independently via discrete uniform distribution over $\{0, 1, \ldots, 21\}$ with probability $p = 1/22$ per substrate.

Cat_Idx (Catalyst Identity, discrete uniform $p = 1/4 = 0.25$ each):

- `0`: Molybdenum Oxide Heterogeneous Nanocatalyst (Base_Cat = 55)

- `1`: Silver-Copper Polyoxometalate (Base_Cat = 45)
- `2`: Supported Palladium Nanoparticles (Base_Cat = 40)
- `3`: TEMPO/Cu(I) Homogeneous System (Base_Cat = 30)

Ox_Idx (Oxidant Identity, discrete uniform $p = 1/3$ each):

- `0`: Molecular Oxygen (O_2, 1 atm) (Base_Ox = 30)
- `1`: Aqueous Hydrogen Peroxide (H_2O_2, 200 mol%) (Base_Ox = 20)
- `2`: Sodium Hypochlorite (NaOCl) (Base_Ox = 10)

S3. Data Generating Process, Oracle Definition, and Decoy Distributions

Twenty independent datasets of $N = 5,000$ reactions were generated by initializing `numpy.random.seed(seed)` for seed $\in \{42, 43, \ldots, 61\}$. Continuous operational and synthetic decoy features were sampled independently for each reaction:

- **Temp:** $\sim \mathcal{U}(25, 110)$°C
- **Decoy_Cat_Load:** $\sim \mathcal{U}(1, 10)$ mol%
- **Decoy_Rot_Bonds:** $\sim \mathcal{U}(0, 10)$ (continuous uniform synthetic noise)
- **Decoy_TPSA:** $\sim \mathcal{U}(20, 120)$ Å^2 (continuous uniform synthetic noise)

Observed yields Y (`Target_Yield`) were computed via Equations 1 and 2 in the main text with $\varepsilon \sim \mathcal{N}(0, 5^2)$ percentage points. The empirical oracle prediction for reaction i is the clamped noiseless expectation $\hat{Y}_{\text{oracle},i} = \min(100, \max(0, \text{Base_Cat}_i + \text{Base_Ox}_i - 0.05M_i + 0.15(T_i - 25)))$, evaluated against the clamped target Y_i.

S4. Green Performance Index (GPI) Exact Calculations

On the **Seed 42 reference dataset** (`synthetic_benchmark_dataset.csv`), the illustrative GPI combines the **observed, noisy, clamped yield (Y, `Target_Yield`)**, selectivity, and sustainability:

$$\text{Raw GPI} = (0.40 \times \text{Target_Yield}) + (0.25 \times \text{Selectivity_Score}) + (0.35 \times \text{Sustainability_Score}) \tag{S1}$$

Sustainability_Score (points):

- O_2 Base = 95, H_2O_2 Base = 85, NaOCl Base = 50
- Sustainability_Score = Base $- 0.5 \times$ (Temp $- 25$)

Selectivity_Score (percentage points): Additive Gaussian noise $\varepsilon_{\text{sel}} \sim \mathcal{N}(0, 2^2)$ (standard deviation of 2 selectivity percentage points, unclipped) is added to each catalyst's baseline:

- Mo-Heterogeneous (`Cat_Idx = 0`): $85 + \mathcal{N}(0, 2^2)$
- Ag-Cu Polyoxometalate (`Cat_Idx = 1`): $75 + \mathcal{N}(0, 2^2)$

- Pd-Nanoparticles (`Cat_Idx = 2`): $70 + \mathcal{N}(0, 2^2)$
- TEMPO/Cu(I) Homogeneous (`Cat_Idx = 3`): $60 + \mathcal{N}(0, 2^2)$

Affine Rescaling: An affine transformation maps Raw GPI for reporting in Table 3 without additional clipping:

$$\text{Norm_GPI} = \left(\frac{\text{Raw GPI}}{100}\right) \times 4 + 1 \tag{S2}$$

S5. Machine Learning Configuration and Fold-Specific Scaling

Fold-Specific Data Preprocessing: Categorical parameters (`Cat_Idx`, `Ox_Idx`) were encoded using One-Hot Encoding. Inside the 5-fold cross-validation loop, a fresh `StandardScaler` instance was fitted exclusively on the continuous columns (`ExactMolWt`, `Temp`, and the three decoy columns when included) of each fold's training split (`X_tr`) and applied to transform that fold's held-out test split (`X_te`).

Cross-Validation and Metric Aggregation: Random validation used `KFold(n_splits=5, shuffle=True, random_state=seed)` synchronized with each dataset's seed. Grouped validation used `GroupKFold(n_splits=5)` grouped by `Substrate_Idx`. Within each seed, MAE and R^2 were computed on each of the 5 test folds and averaged to obtain a single seed-level score. Tables 1 and 2 report the mean and sample standard deviation (`ddof=1`) across the 20 seed-level scores.

Fixed Model Hyperparameters:

- **Linear Regression:** `LinearRegression()` (default OLS).
- **XGBoost Regressor:** `n_estimators=50`, `max_depth=3`, `learning_rate=0.1`, `random_state=seed`.
- **Random Forest Regressor:** `n_estimators=50`, `max_depth=5`, `random_state=seed`.
- **Support Vector Regression (SVR):** `kernel='rbf'`, `C=1.0`, `epsilon=0.1` (untuned).

S6. Computational Environment

All data generation, cross-validation, and SHAP analyses were executed in Python 3.10+ (Google Colaboratory Linux environment) using `numpy` (≥ 1.25), `pandas` (≥ 2.0), `scikit-learn` (≥ 1.2), `xgboost` (≥ 2.0), `shap` (≥ 0.45), and `rdkit` (v2023.09.5).